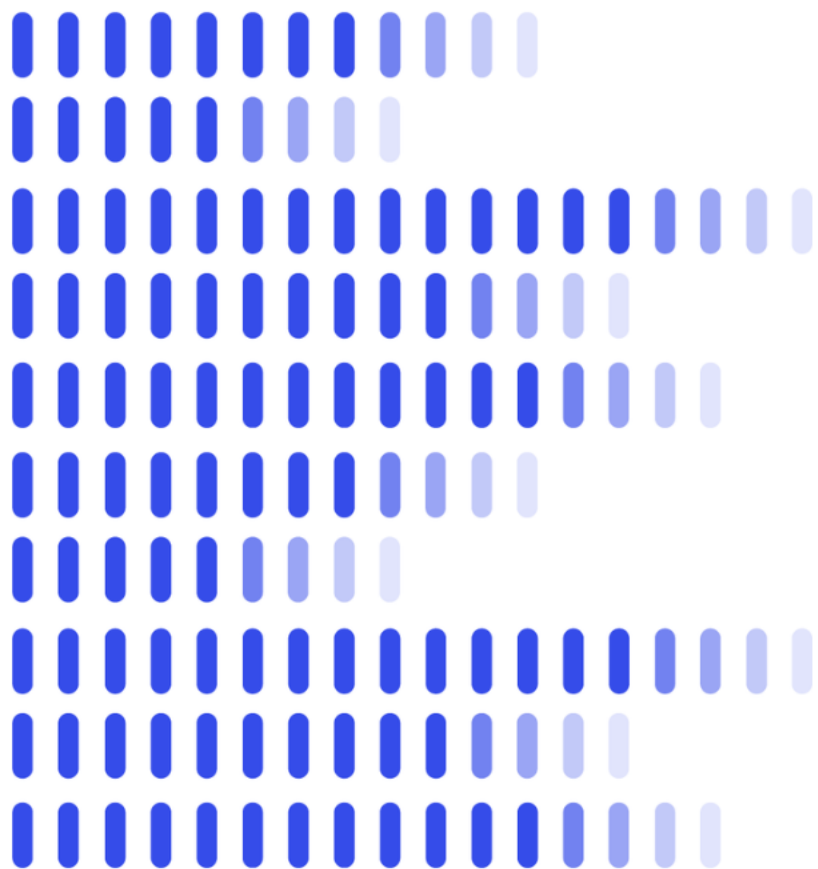
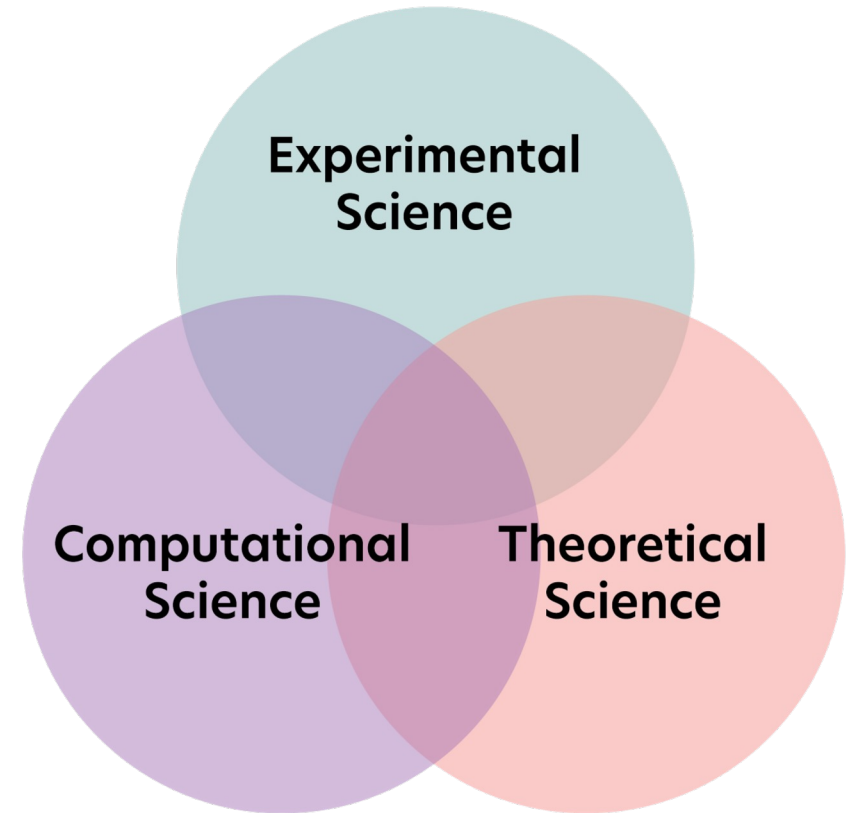




Minho Advanced Computing Center

The Three Paradigms of Science

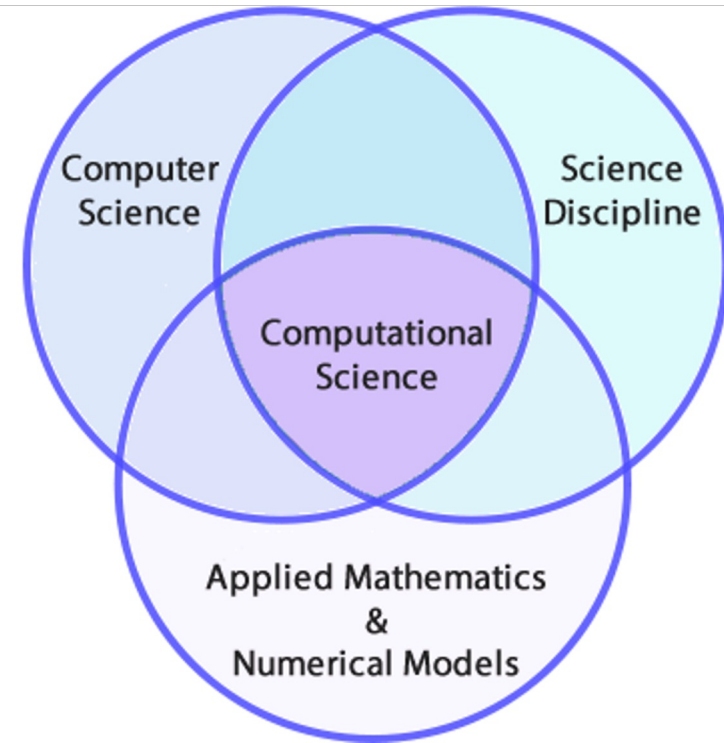
- In scientific research, various methods are used to find explanations for natural phenomena and make predictions about their behavior
- Traditional
 - Experimental
 - Theoretical
- Computational



Source: Edukamu Elements of supercomputing course

Computational Science/Scientific Computing

- There are limits to what can be done with pen and paper
- What kind of experiments can be set up.
- Typically complex mathematical models describing the problem
- Are solved using numerical algorithms
- Implemented as computer programs and finally run on high-performance computers

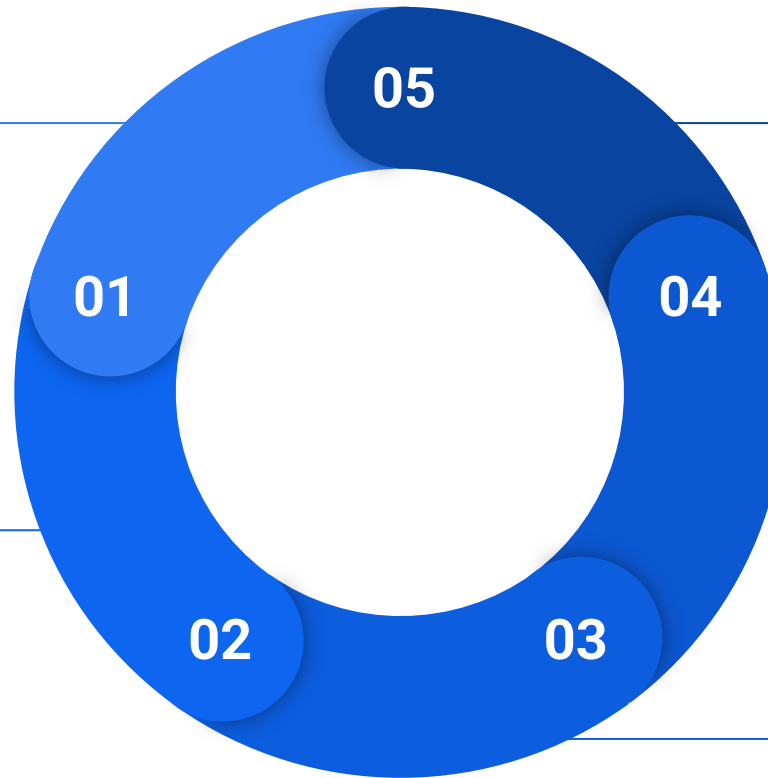


Source: <https://computingstudy.wordpress.com/scientific-computing/>

Computational Science Steps

Theoretical Model
Develop the scientific domain theoretical models.

Computational Model
Choose or implement computational solvers for the theoretical model



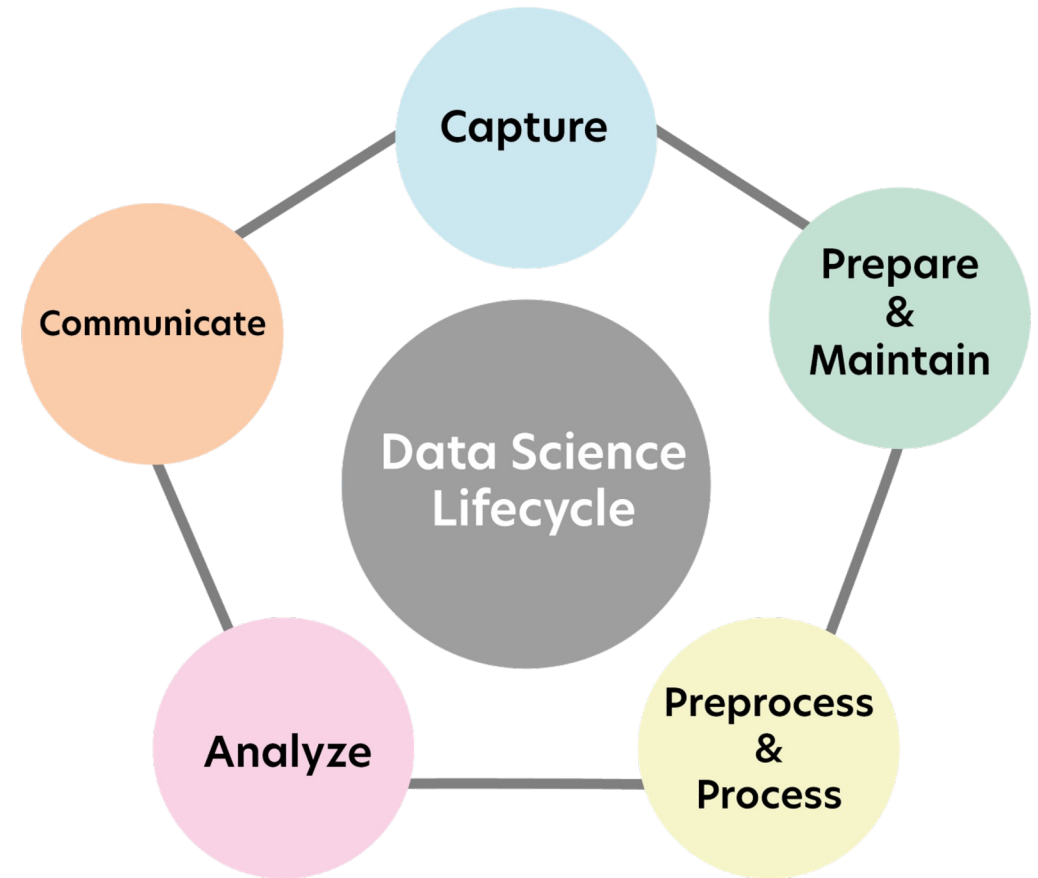
Validation / Verification
Evaluate the computational results, run physical experiments or check against theoretical model.

Result gathering
Run the large scale computational experiments and gather the results

Code Development
Implement the computational model in some language ensuring that it is able to explore the underlying resources

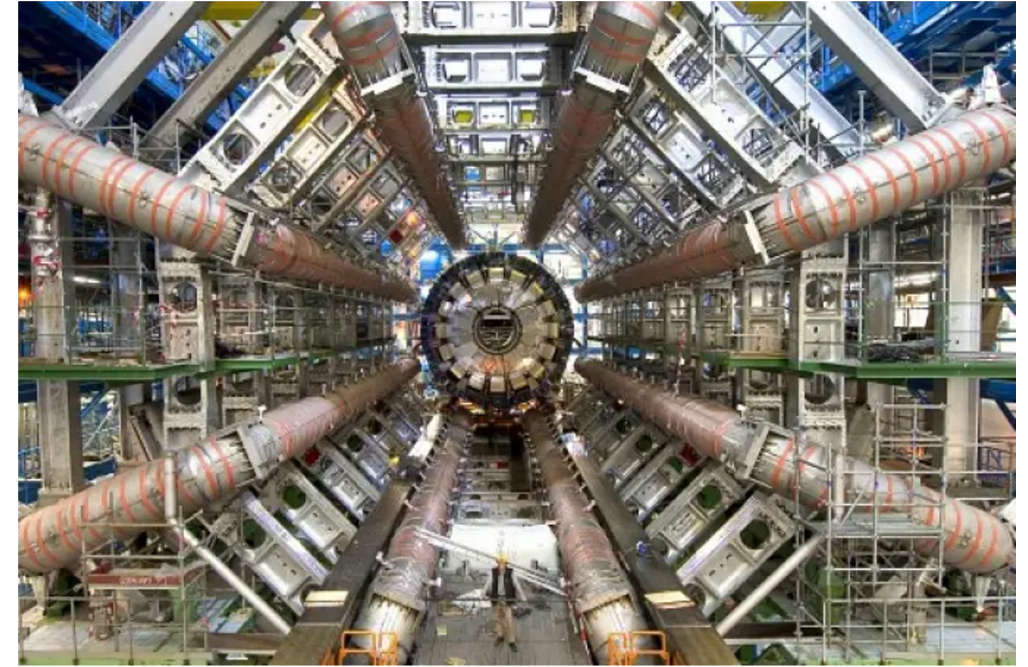
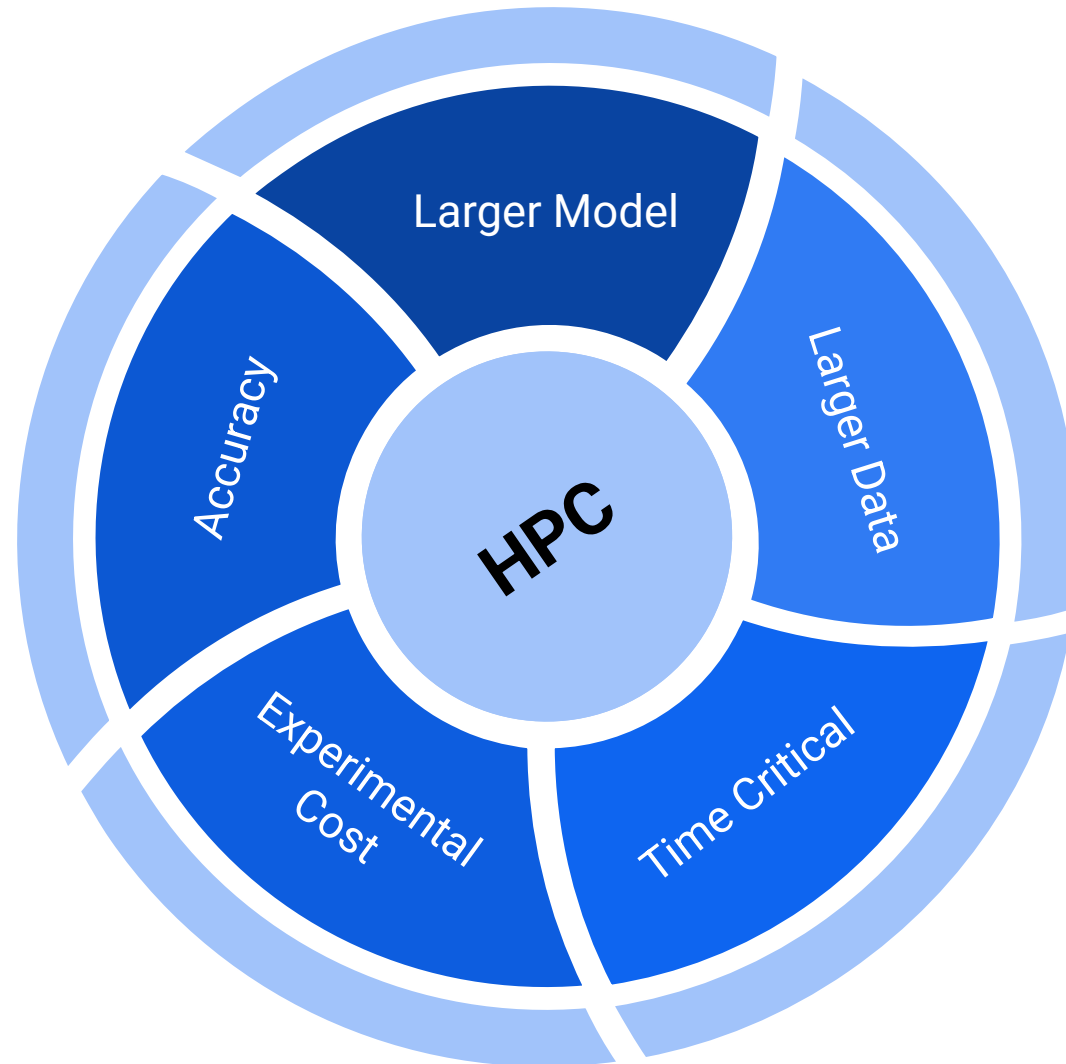
Data Science

- Paradigm shift
 - Traditional : Theoretical -> Data
 - Data Science : Data -> Theoretical Model
- Uses various mathematical methods and algorithms to extract knowledge and insight from data



Source: Edukamu Elements of supercomputing course

Need for HPC



Source: CERN ATLAS



Source: LIGO Observatory

Misconceptions on HPC

“My program will run faster”

“A super-computer can run the same software as my desktop”

“I can run my experiment whenever and however I want”



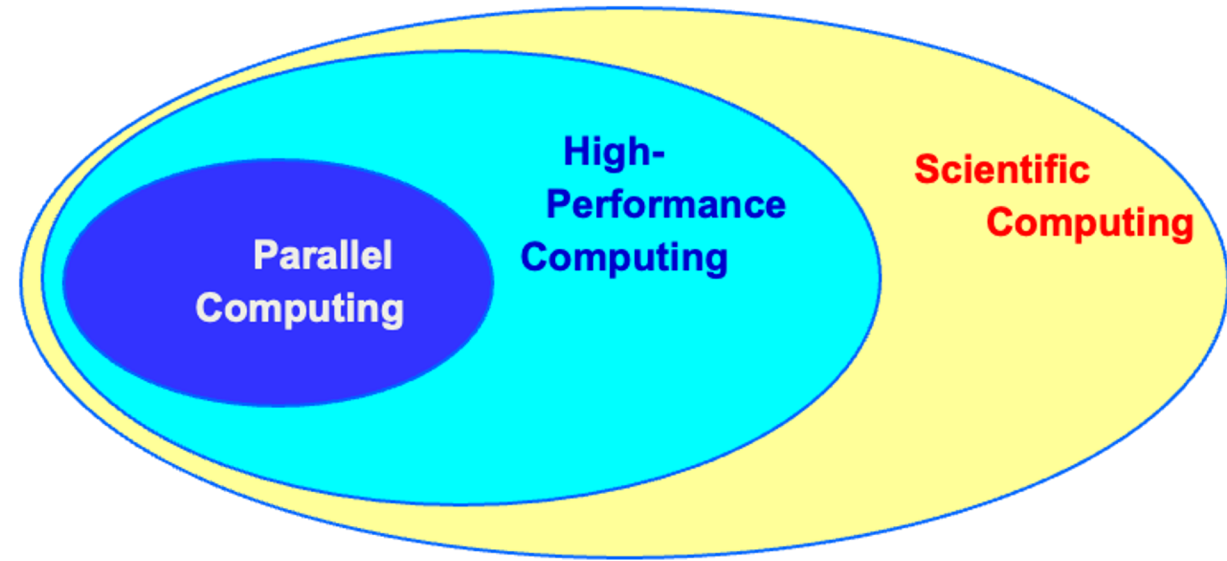
Supercomputer

- User experience
 - Multiuser
 - Optimized software
- Compute power
 - Many CPUs
 - Specialized HW
 - Low latency/High bandwidth network
 - Large memory
 - Efficient I/O



Highly Parallel Environment

- Concurrent use of multiple processors to process data
- Shared-memory architecture
 - e.g. : OpenMP
- Distributed-memory architecture
 - e.g. : MPI



Highly Parallel Environment - Advice

Language

Choose the language that you are comfortable with, but make sure it has libraries that allow you to scale up.

Libraries

There are a collection of libraries that already implement the solvers you need, but make sure that they are maintained regularly, have a large community and scale up.

Code

Minimize your code, but if you can ensure that the code is modular to implement features and enable scalability as you improve the code.

Working With a Supercomputer

COPY DATA

The space on a HPC Cluster is not infinite

Copy the data into the cluster storage as needed

PREPARATION

Most of time your model does not come out of the box and need compiling

You need to specify a script to automate the execution step

EXECUTION

Submit you computational pipeline to the cluster for batch execution

The system will choose when to run based on the resources requested

ANALYSE

Most of the time the resulting data is too large to be copied back as raw data

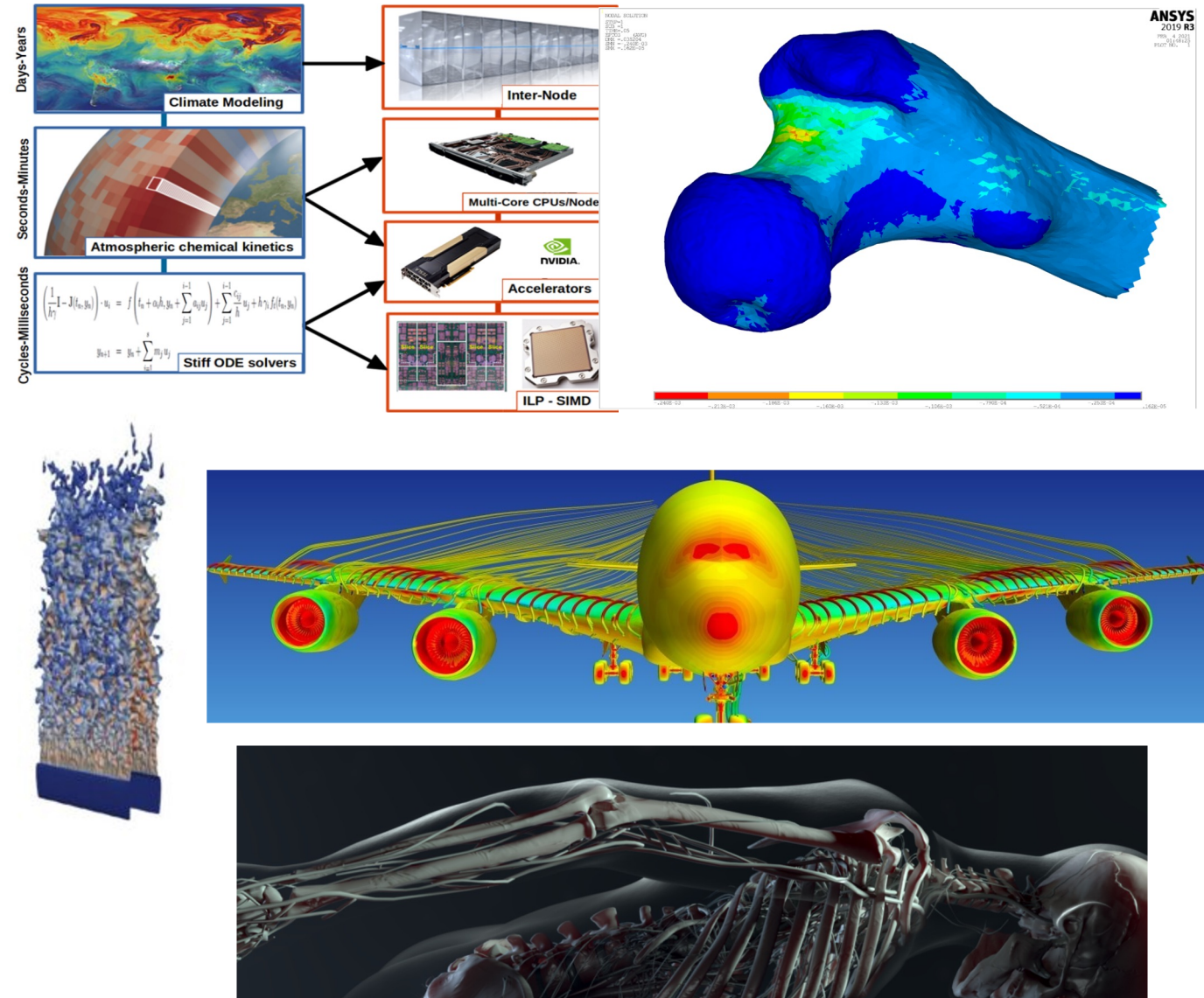
The data can be analysed locally and in some system allows interactive sessions for analysis and scientific visualization

RETRIEVE DATA

Once again space is not infinite, make sure you copy the important results back

HPC Impact

- Utilized in many fields of science and research
 - Simulation
 - fuel efficiency of aircraft
 - better wind turbines
 - nanoparticles collision
 - aerosols dispersion
 - climate models
 - Natural language processing
 - Medicine
 - drug discovery
 - how the human brain works
 - cancer diagnosis

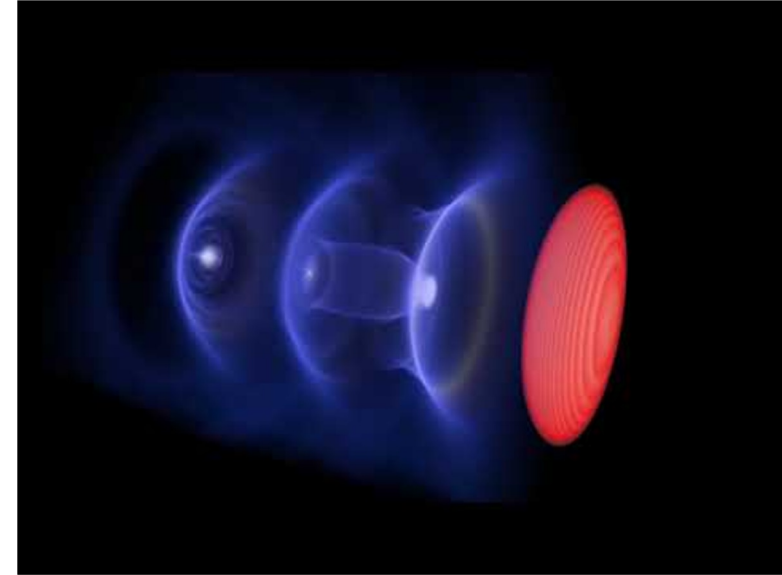


Source:

Computational Science and HPC

PHYSICS

Molecular modeling, electronic circuit design, protein folding, atmospheric science, aerodynamic design and testing, material science



Source:
Smilei

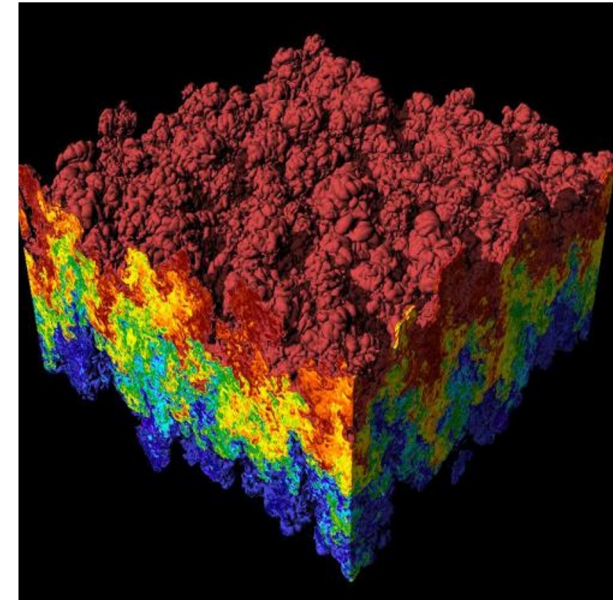
COMSOL
MULTIPHYSICS®



Smilei)

Osiris

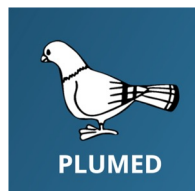
b-initio
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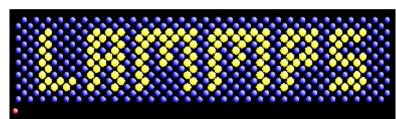
Computational science and HPC

CHEMISTRY

Calculate the structures and properties of molecules and solids, applying these programs to real chemical problems



AMBER MD



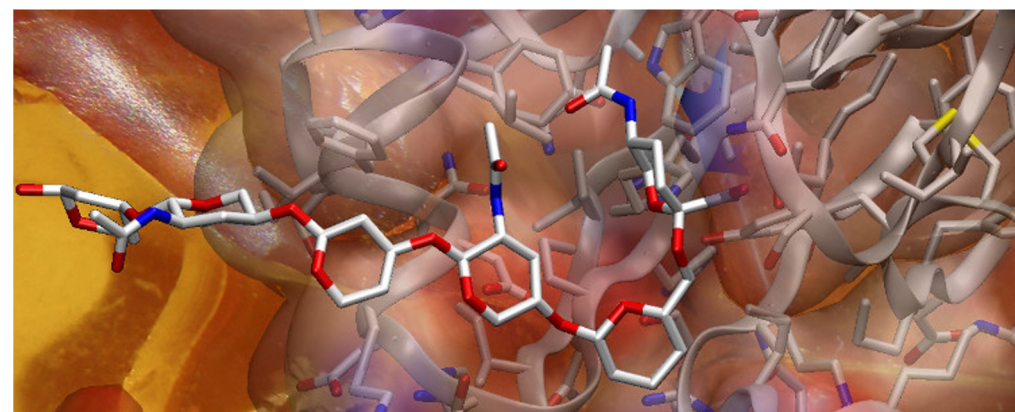
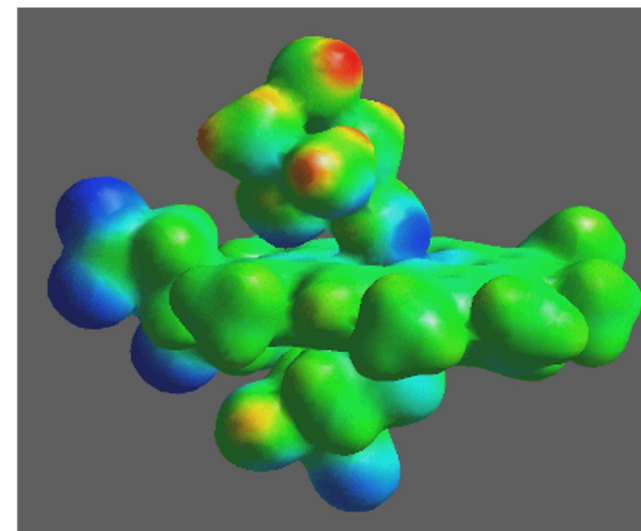
GAMMESS



GROMACS
FAST. FLEXIBLE. FREE.



NAMD
Scalable Molecular Dynamics

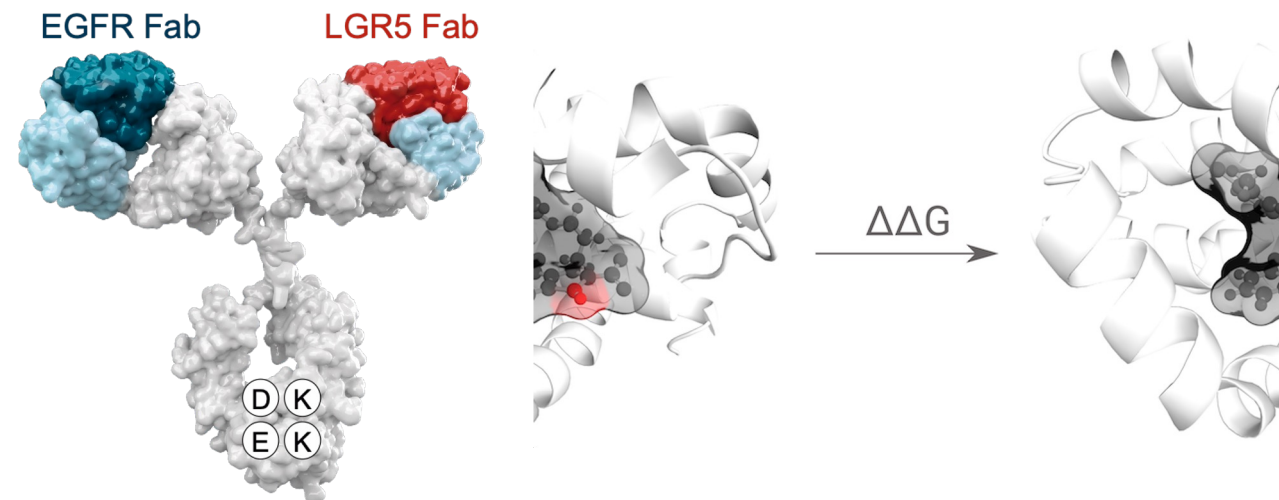
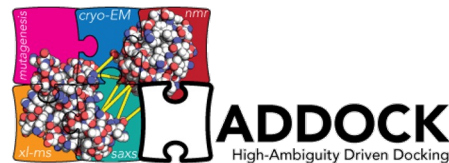


Source: Amber
MD

Computational science and HPC

BIOINFORMATICS

- Drug discovery, optimization of treatment schemes
- Gene expression and regulation
- Personalized Medicine
- Aid in the design and optimization of biotechnological processes
- How proteins interact with each other



Computational science and HPC

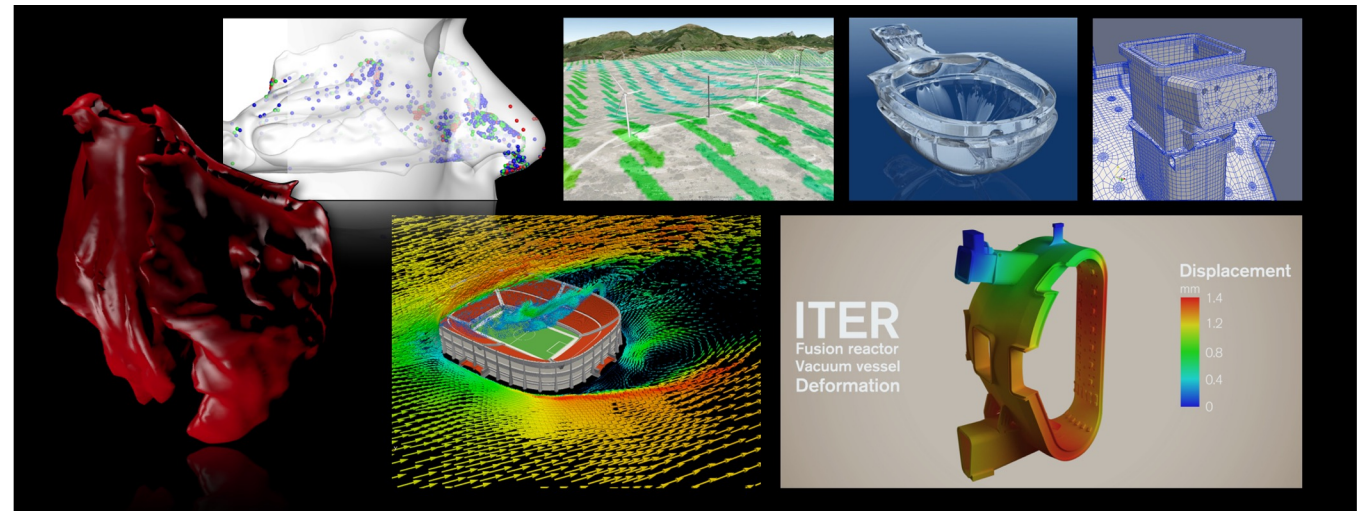
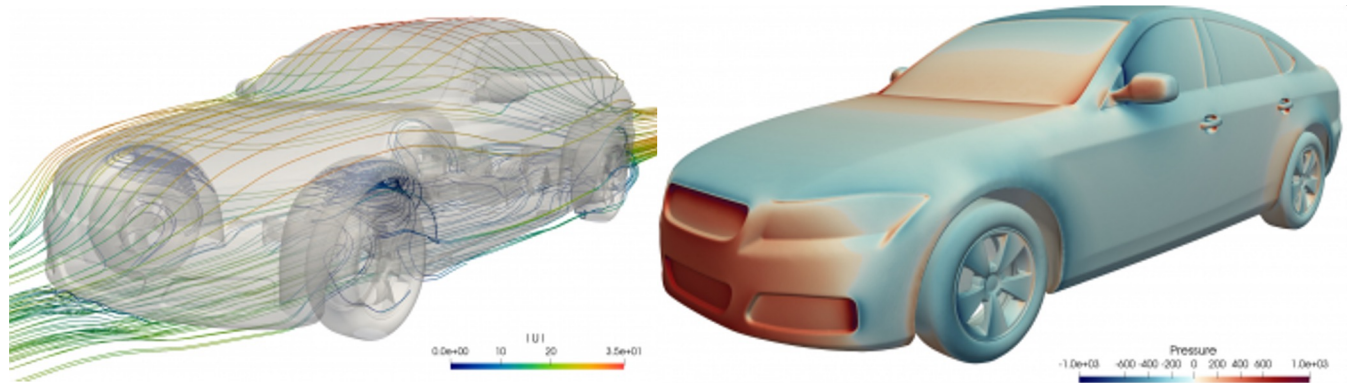
MATERIALS SCIENCE

Vehicle crash simulation,
petroleum reservoir modeling,
biomechanics, glass
manufacturing, and
semiconductor modeling

Open  FOAM®

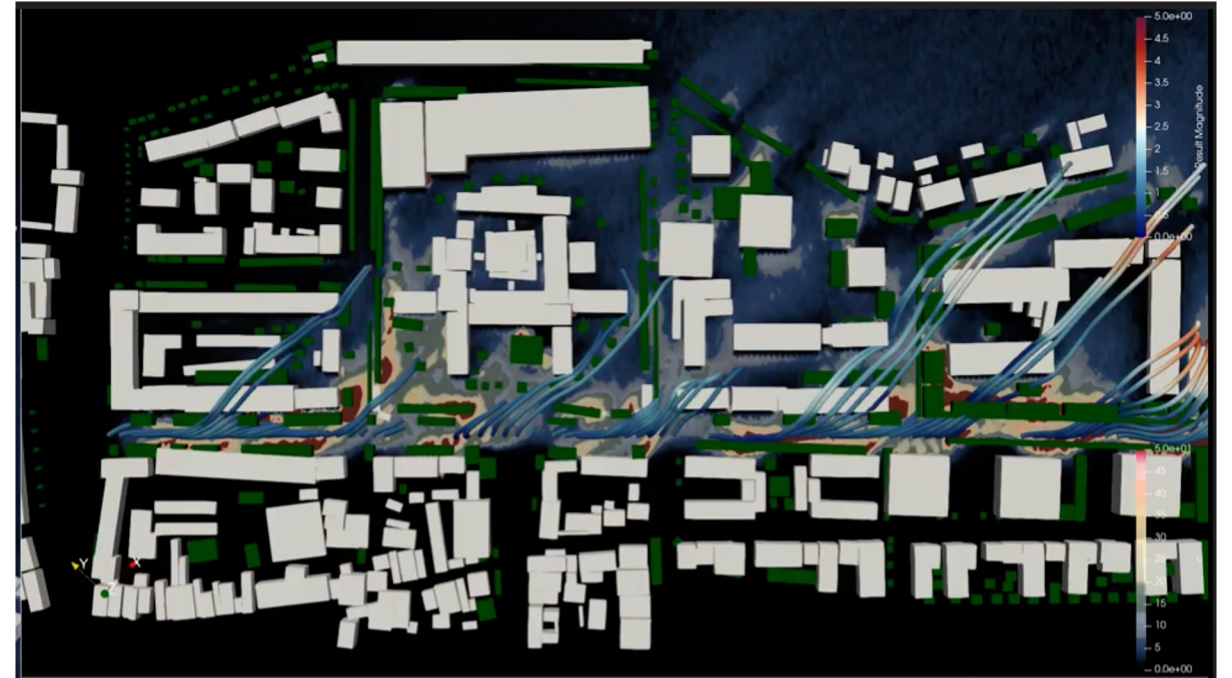
 **Ansys**

Alya
HPCM



Visualization

- Explore large dataset
- Visual comparison with experimental results
- Scientific communication



MACC

Advanced computing research and innovation

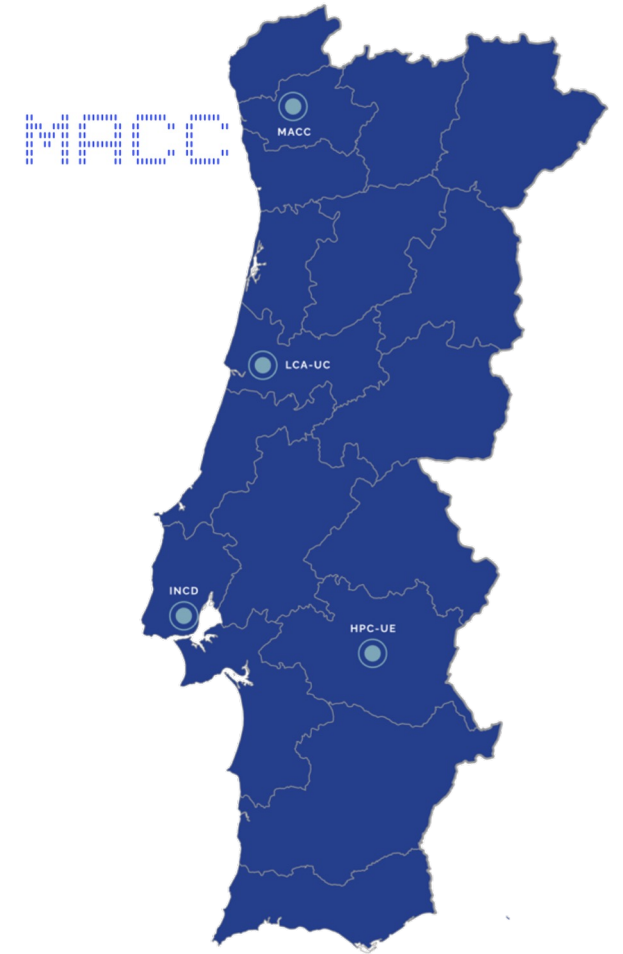
Support and foster research and innovation on the codesign of HPC systems from their implementation to optimization and exploitation, a major priority towards Europe's HPC and EPI initiatives.

Computational science and engineering, and Artificial intelligence research

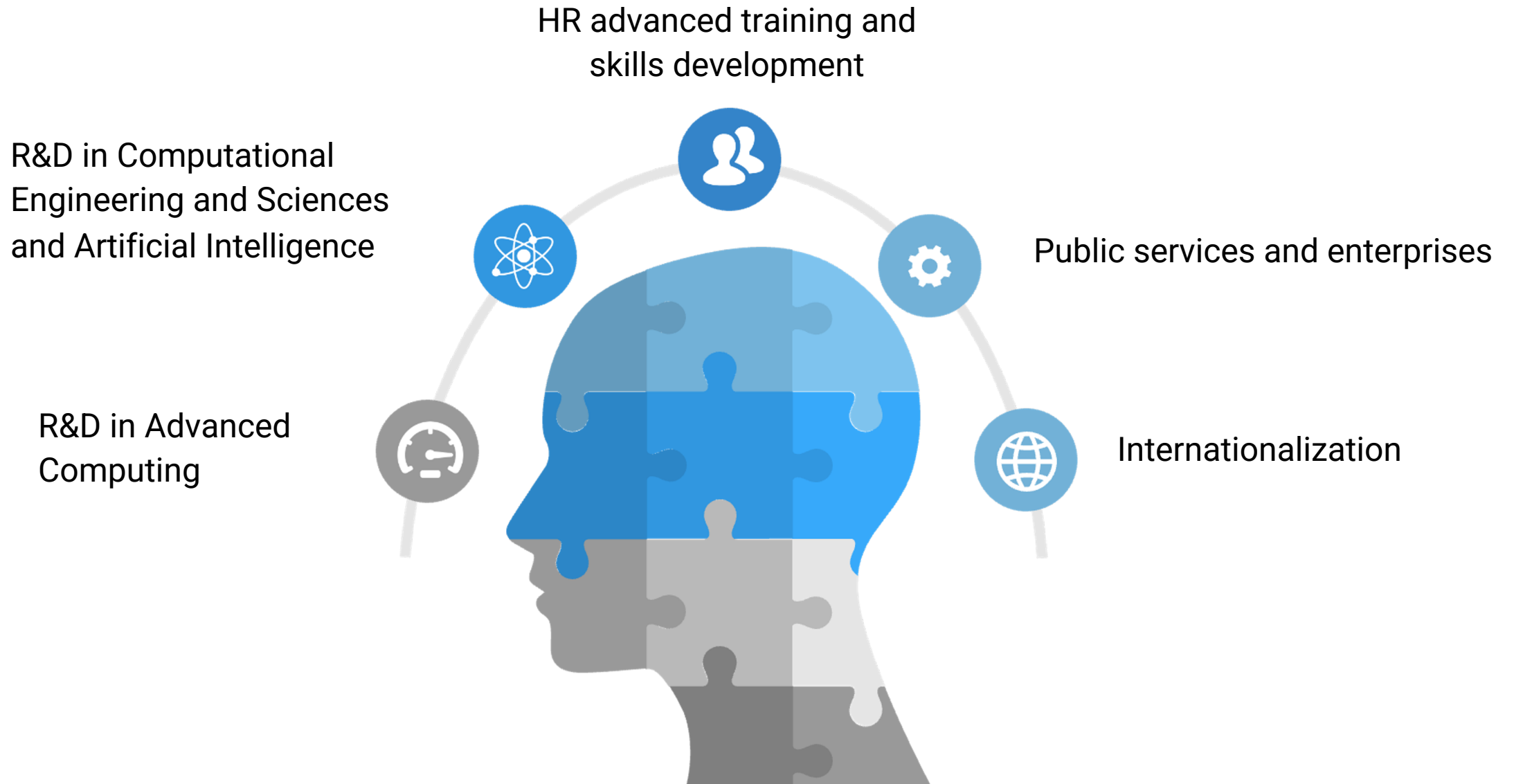
Foster the creation, expansion and consolidation of research and innovation teams on national and European high priority scientific domains requiring large scale digital simulations and data intensive systems and applications.

Public services and enterprises

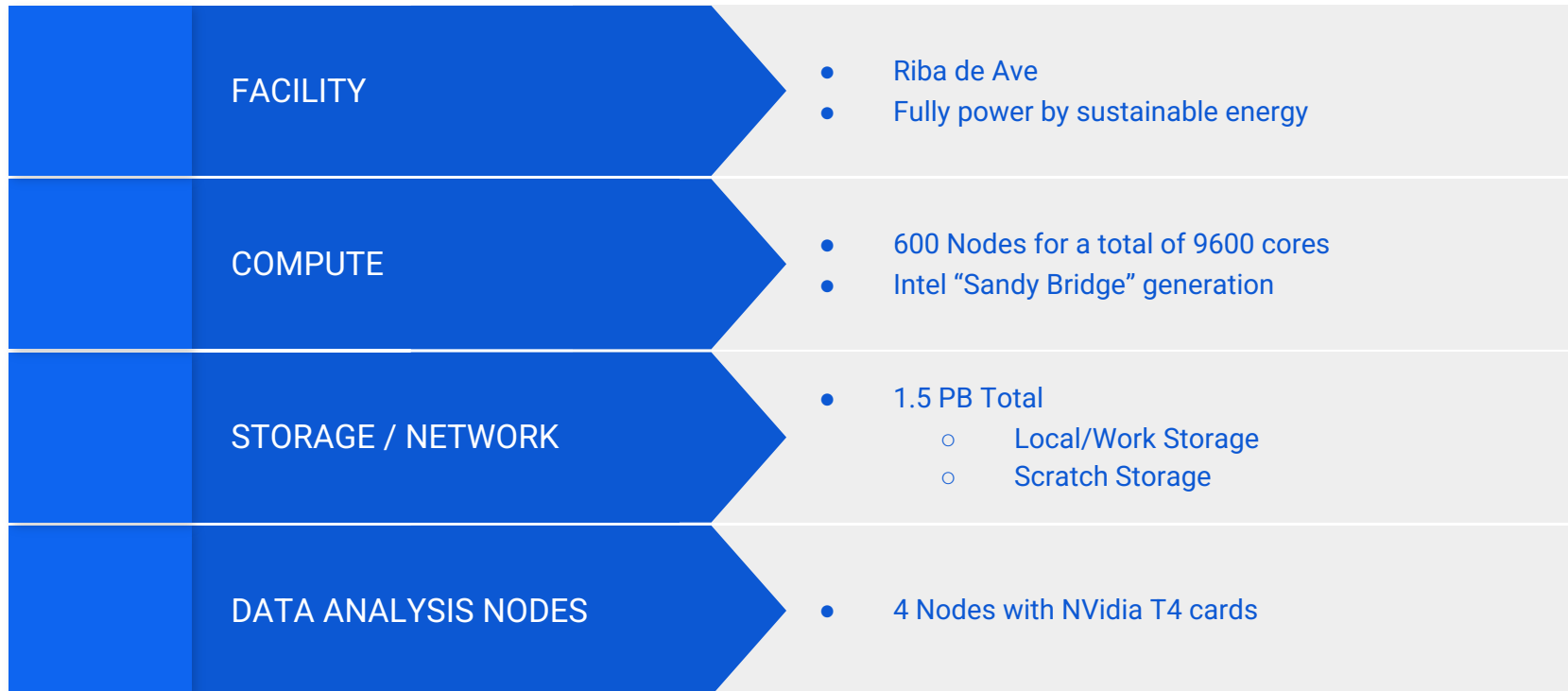
Offer of advanced computing resources and consultancy to science and higher education institutions, public administration, academic entrepreneurship and enterprises for the curation, management and processing of big data and machine learning workload.



Objectives



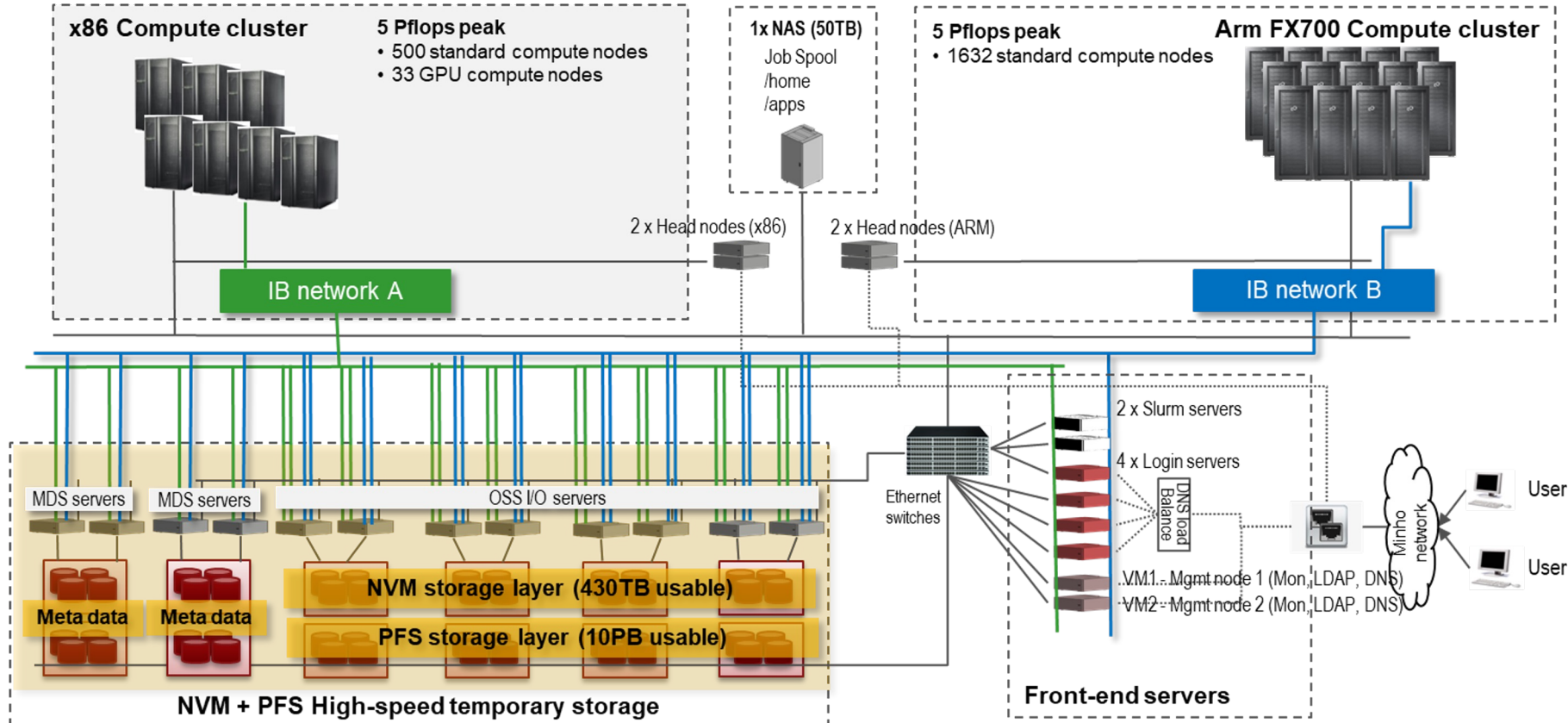
Bob



Deucalion

- Portugal application to a EuroHPC petascale machine
- Two general purpose systems with overlapping but dissimilar goals
 - X86- Mostly CPU based nodes and few GPU equipped nodes
 - Arm V8 with Scalable Vector Extension (SVE)

Overall Architecture



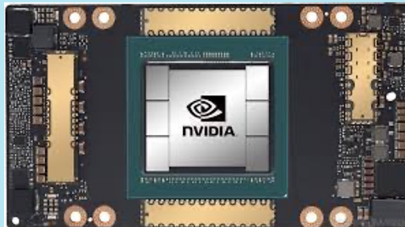
Deucalion Compute Nodes



Compute nodes – 1632
Cores Number – 78,336
Memory Capacity – 52 TB
Rpeak – 5.013 PFlops

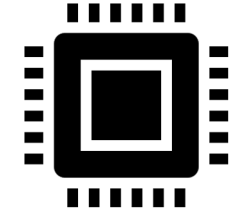


Compute nodes – 500
Cores Number – 64,000
Memory Capacity – 128 TB
Rpeak – 2.304 PFlops



Compute nodes – 33
CPU Cores Number – 4.224
Memory Capacity – 16 TB
GPU Memory – 8 TB
Rpeak CPU – 152,064 GFlops
Rpeak GPU – 2.572 PFlops

10 PFlops



2 clusters / 3 partitions

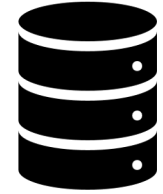
- The ARM cluster is based on the Fujitsu A64FX processor with high levels of performance with low energy consumption
- The x86 cluster with AMD EPYC highly efficient processor with very good HPL efficiency and excellent energy
- The accelerator nodes have Ampere GPU from NVIDIA

Deucalion Storage

High Speed Storage

Metadata and Hot Pools NVMe – 430 TB usable
HDD Datapools – 10 PB usable
MDS Servers – 8 Nodes
OSS I/O Servers – 32 Nodes
Filesystem – Lustre PFS

Aggregated Performance
- 340GB/s in reads, 260GB/s in writes



- Building block architecture High speed storage with both an NVME tier and a traditional PFS disk-based tier



NAS Storage

Total SSD net Capacity – 50 TB
Data Modules – 2 for redundancy
Connection type – 8x10 GbE

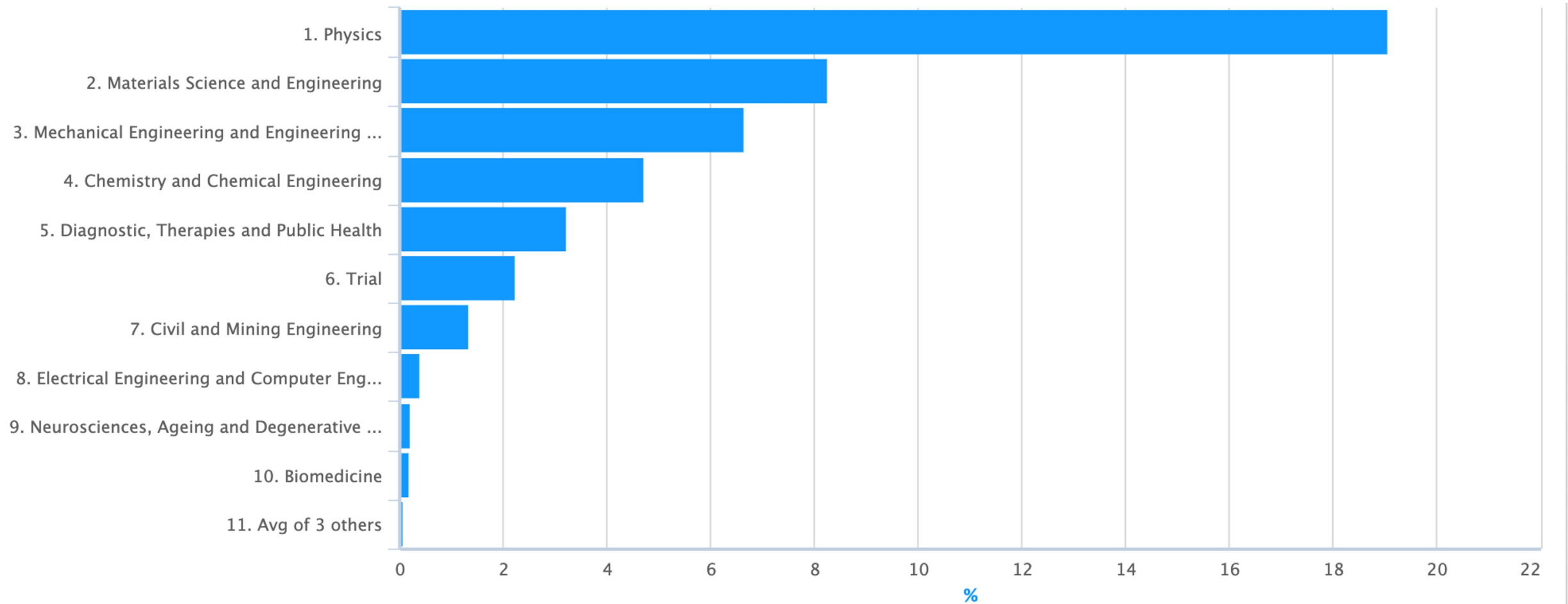
- NAS subsystem for the purposes of the user homes (/home), application sources and binaries (/apps), as well as Job Spool and node images. Provides visibility of data across all clusters and servers.

Deucalion for AI

- 33 nodes with 4 NVIDIA Ampere A100
- 1632 nodes with Fujitsu A64FX processor with ARM-SVE
 - Single-precision and half-precision floating-point calculations, as well as 8-bit and 16-bit calculations with 512-bit wide SIMD, with high throughput
 - Allow for machine learning applications (both training and inference)
- Resnet-50 training time

Device	Training Time (s)	Nodes/cores	Power consupction (kWh)
CPU X86 AMD Rome	20000	500 / 64000	778
CPU ARM A64FX	376,380	16 / 768	323,90
GPU Nvidia Ampere	190	33 / 228096	2,35

Current Usage by Area



Current Use Cases

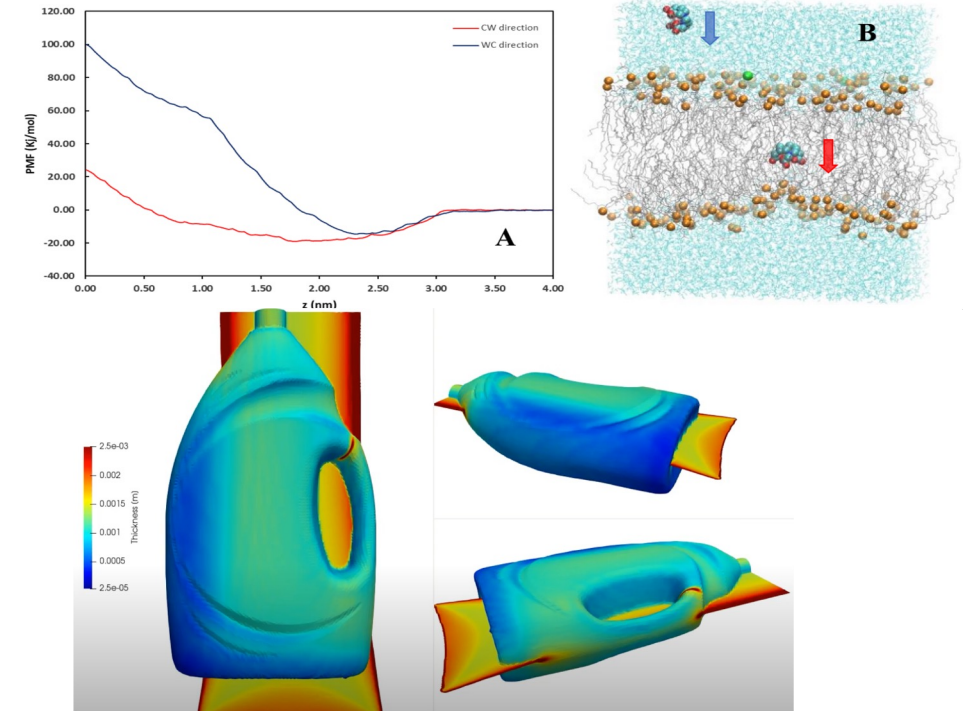
OpenFOAM for industrial simulation namely covering polymer and metal processing, health, processing and energy areas

Direct numerical simulations (DNS) of turbulent jets and wakes, for investigating the physics of turbulence dynamics

GROMACS for Molecular Dynamics (MD) simulations of a key protein in explicit solvent and in a membrane system

Incorporation of water in a porous material using classical molecular dynamics simulations

Turbine siting simulation of air flow over terrain



VADIS – GEMAC – DAO – U.Aveiro

Compute Pollutant Concentration in an Urban Environment

Code modernization

- Extremely old version of Fortran → Recode in C++'17
- Fluid solver uses an iterative solver similar to a Jacobian Relaxation
 - Use OpenMP as a parallel framework

Dispersion solver uses an approach similar to marching cubes

- When a particle is moved from one grid cell to the next based on its momentum and the fluids momentum
- When particle enters a building march the particle until it exist the building
 - Use a ray tracing algorithm to solve particle building collision
 - Use a task-based programming environment

In-situ visualization

- Computational steering
- Results checking
- Avoid expensive I/O

